KIRUBEL YONAS

Systems Engineer & Security Researcher
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PROFESSIONAL SUMMARY

Low-level Linux systems engineer and security researcher specializing in C++20/C development, DevSecOps automation, and binary security. Practical experience developing dynamic runtime analysis tools, binary tracing engines, and automated security auditing platforms. Proven track record in vulnerability research, AFL++ coverage-guided fuzzing, and security tool construction.

TECHNICAL SKILLS

- **Languages:** C++, C, Python, Rust, Go, TypeScript, JavaScript, Kotlin, Bash, SQL
- **Systems & Low-Level:** Linux Architecture (Arch/Debian), `ptrace`, ELF Instrumentation, System Call Interception
- **DevSecOps & Cloud:** Docker, CI/CD Security Pipelines, Git, SAST Engine Construction
- **Offensive Security & Research:** AFL++ Fuzzing, Ghidra, Burp Suite, Wireshark, Metasploit, Reverse Engineering

EXPERIENCE

LaloDev — Software Engineer

June 2025 – November 2025 | Addis Ababa, Ethiopia

- Modernized backend services within an agile SDLC, focusing on execution speed, resource efficiency, and memory safety.
- Built custom Python automation scripts to integrate automated security checks and secure deployment patterns into CI/CD pipelines.
- Conducted threat modeling and code reviews to harden core application logic against common OWASP vulnerabilities.

KEY TECHNICAL PROJECTS

dSBOM-engine | Runtime Software Bill of Materials (SBOM) Engine

- Architected a C++20 engine that intercepts kernel system call boundaries using `ptrace(2)` and `PTRACE_SYSCALL` to map runtime dependencies dynamically from the Linux VFS.

Telegram Animation Engine | Media Fuzzing & Memory Safety Research

- Engineered an AFL++ fuzzing harness targeting the `rlottie` animation library used in Telegram's sticker engine.
- Uncovered memory corruption edge cases and input-validation flaws using corpus minimization and custom dictionary techniques.

ZeroShadow | Dynamic ELF Binary Tracing Engine

- Built a high-performance C++17 execution tracing engine using `ptrace` hooks to inspect ELF binary execution paths for dynamic malware analysis.

SAUR | Open-Source Security Utilities

- Created a lightweight YARA rules engine (SAUR) and static analysis scanner to detect suspicious patterns and malicious build scripts in AUR package caches.

Threat-Matrix | Edge Threat Intelligence Platform

- Designed a real-time threat intelligence pipeline and Android client (TypeScript/Kotlin) to ingest, process, and stream live global CVE alerts and security telemetry.

EDUCATION & CREDENTIALS

- **High School Diploma** — Radical Academy *(May 2026)*
- **Hackathons** — 2x Winner, ALX Engineering Hackathon
- **Certifications:**
  - Hack The Box (HTB) — *Password Attacks Specialist*
  - freeCodeCamp — *Data Structures & Algorithms*, *Responsive Web Design*
  - SoloLearn — *C++, JavaScript, SQL Certifications*